

Homework 6

MAE 384

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Q1

Problem Statement: I am to solve the diffusion equation through PDEs to solve for the concentration of pollutants in a long thin air duct.

Solution:

$$\frac{\partial c}{\partial t} - \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right) = 0$$

$$\frac{\partial c}{\partial x}(0, t) = 0$$

$$\frac{\partial c}{\partial t} - D \frac{\partial^2 c}{\partial x^2} = 0$$

$$\frac{\partial c}{\partial x}(L, t) = 0$$

$$c(x, t) = f(x) g(t)$$

$$g' f - D f'' g = 0$$

$$\frac{g'}{Dg} = \frac{f''}{f} = -\lambda^2$$

$$f'' + \lambda^2 f = 0 \quad f(x) = A_1 \cos \lambda x + A_2 \sin \lambda x$$

$$g' + \lambda^2 Dg = 0 \quad g(t) = e^{-\lambda^2 D t}$$

BC:

$$f'(0) = -\lambda A_1 \sin(\lambda \cdot 0) + \lambda A_2 \cos(\lambda \cdot 0)$$

$$f'(0) = \lambda A_2 = 0$$

$$A_2 = 0$$

$$f'(L) = -\lambda A_1 \sin(\lambda L) = 0$$

$$\lambda L = n\pi$$

$$\lambda = \frac{n\pi}{L}$$

$$f_n(x) = A_n \cos\left(\frac{n\pi x}{L}\right)$$

$$c(x, t) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) e^{-\lambda^2 D t}$$

→

$$C(x,t) = \sum_{n=0}^{\infty} A_n \cos(\lambda_n x) e^{-\lambda_n^2 D t} \quad \lambda = \frac{n\pi}{L}$$

I.C.

$$C(x,0) = C_0 \left(1 - \cos\left(\frac{2\pi x}{L}\right)\right)$$

$$= \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) = A_0 + A_1 \cos\left(\frac{\pi x}{L}\right) + A_2 \cos\left(\frac{2\pi x}{L}\right) \dots$$

$$= \left[C_0 \left[1 - \cos\left(\frac{2\pi x}{L}\right) e^{-\frac{4\pi^2}{L^2} D t} \right] \right]$$

2

Steady state

$$\lim_{t \rightarrow \infty} \left[C_0 \left[1 - \cos\left(\frac{2\pi x}{L}\right) e^{-\frac{4\pi^2}{L^2} D t} \right] \right] = C_0 = 3$$

Steady state is over a long period of time so we take the limit as $t \rightarrow \infty$ to get the answer

Results: I solved the diffusion equation through separation of variables and then solving each ODE then recombine them. I ended up with an equation to solve for the concentration of pollutants along the pipe at distance 'x'.

Q2

Problem Statement: I wrote a matlab program to numerically solve for the concentration equation using the explicit method.

Solution: This to take the limit of the solution from Q1 as $t \rightarrow \infty$ to determine the steady state of the concentration. We get C_0 which equals 3 for the steady state for both methods.

$$C(x,t) = \sum_{n=0}^{\infty} A_n \cos(\lambda_n x) e^{-\lambda_n^2 D t} \quad x = \frac{n\pi}{L}$$

I.C.

$$C(x,0) = C_0 \left(1 - \cos\left(\frac{2\pi x}{L}\right)\right)$$

$$= \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) = A_0 + A_1 \cos\left(\frac{\pi x}{L}\right) + A_2 \cos\left(\frac{2\pi x}{L}\right) \dots$$

$$= C_0 \left[1 - \cos\left(\frac{2\pi x}{L}\right) e^{-\frac{4\pi^2}{L^2} D t}\right]$$

2

Steady State

$$\lim_{t \rightarrow \infty} \left[C_0 \left[1 - \cos\left(\frac{2\pi x}{L}\right) e^{-\frac{4\pi^2}{L^2} D t}\right] \right] = C_0 = 3$$

Steady state is over a long period of time so we take the limit as $t \rightarrow \infty$ to get the answer

Explicit Method:

Solution: This code uses Euler's Explicit method by using the finite difference method in a for loop for the length-1 of the grid points of x & t . I inputted the initial condition, and the 2 boundary conditions to ensure that the for loop goes along the conditions I want it to. I then plotted the steady state and concentration over T on the same graph, and then compared the finite difference method to the analytical solution (Or the true solution) from Q1 to determine the accuracy of euler's explicit method.

```

%Givens
L = 10;
tf = 100;
c0 = 3;
DE = 0.008;
dxE = 0.01;
dtE = 0.001;

%Grid
x = 0:dxE:L;
t = 0:dtE:tf;

%Number of grid points
Gx = length(x);
Gt = length(t);

% Courant number
CourantE = DE * dtE/dxE^2;

% Initialize concentration matrix
cE = zeros(Gx, Gt);

% Set initial condition
cE(:, 1) = c0 * (1 - cos(2*pi*x/L));

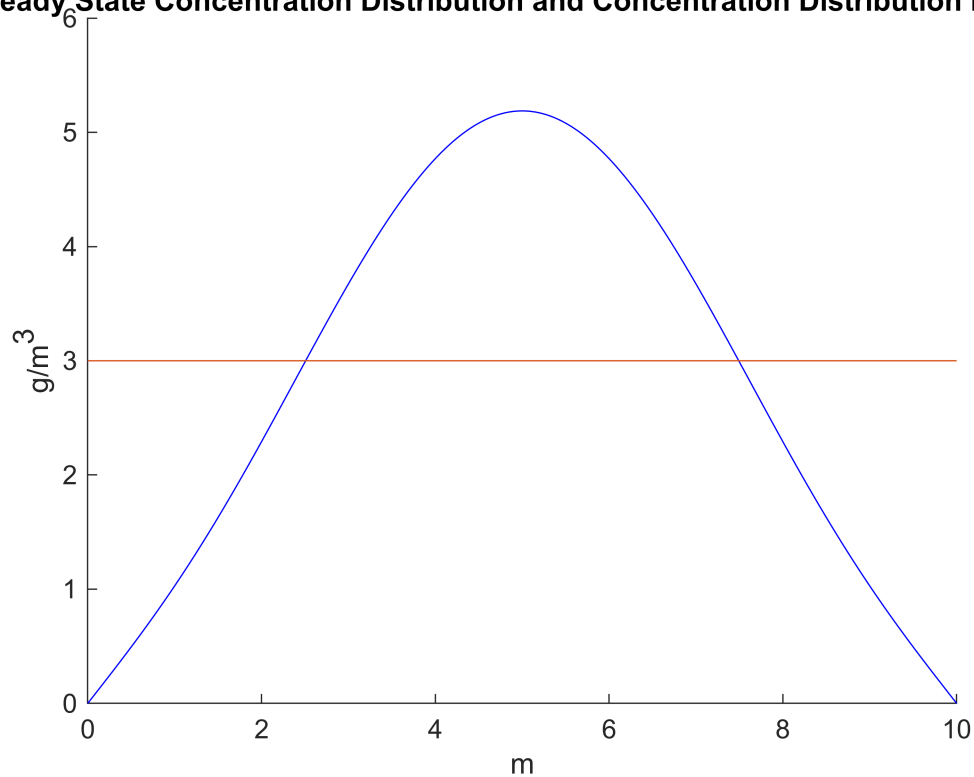
% Apply boundary conditions
cE(1, :) = 0;           % Left boundary
cE(Gx, :) = 0;          % Right boundary

% Perform time stepping
for n = 1:Gt-1
    for i = 2:Gx-1
        % Compute the next time step using the finite difference method
        cE(i, n+1) = cE(i, n) + CE * (cE(i+1, n) - 2 * cE(i, n) + cE(i-1, n));
    end
end

%Concentration Distribution and Steady State Plots
figure;
hold on
plot(xE, cE(:, end), 'b-');
xlabel('m');
ylabel('g/m^3');
%Steady State @ 3 proven through limits on paper
c_steadyE = 3 * ones(GxE, 1);
plot(xE, c_steadyE);
title('Steady State Concentration Distribution and Concentration Distribution Explicit');
hold off

```

Steady State Concentration Distribution and Concentration Distribution Explici

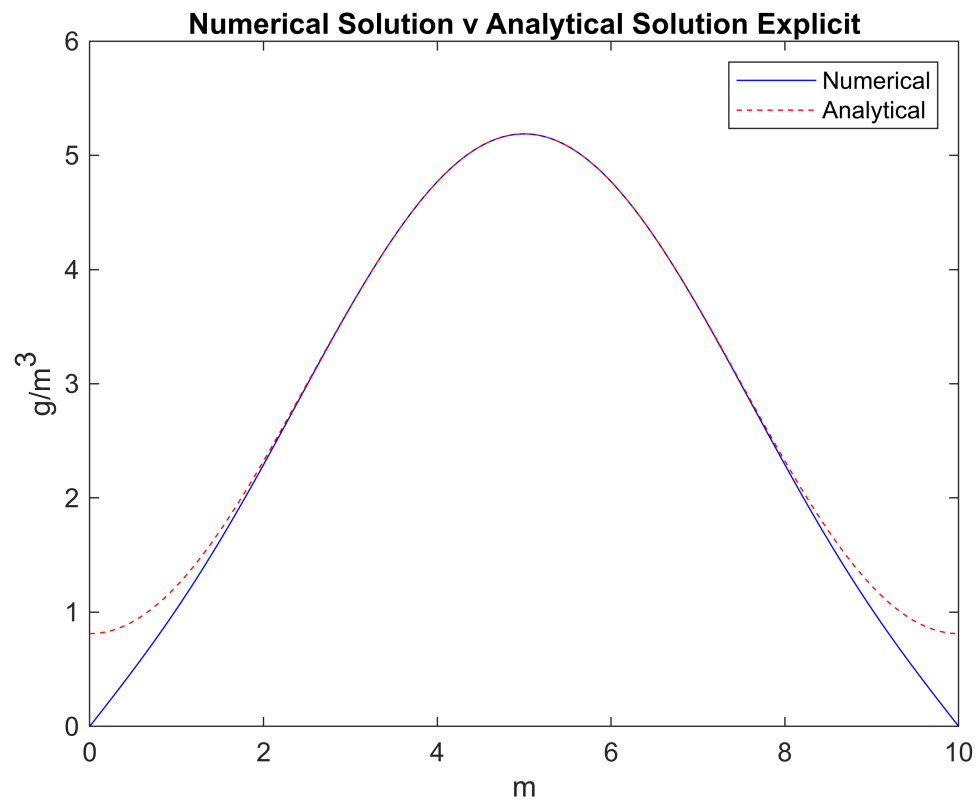


%P1 Analytical Solution

```
canalyticalE = c0 * (1 - cos((2*pi*x)/(L)) .* exp((-4*pi^2)/(L^2)*(DE*tf)));
```

%Plot to compare Numerical and Analytical Solutions

```
figure;
plot(x, cE(:, end), 'b-');
hold on;
plot(x, canalyticalE, 'r--');
xlabel('m');
ylabel('g/m^3');
title('Numerical Solution v Analytical Solution Explicit');
legend('Numerical', 'Analytical');
```



Implicit Method:

Solution: This code uses Euler's Implicit method. I created a matrix of coefficients and then multiplied them by the boundary conditions to create a linear system to solve for the solution of the numerical solution. I did this through a for loop defined by the number of time steps. I then plotted the steady state and concentration over T on the same graph, and then compared the finite difference method to the analytical solution (Or the true solution) from Q1 to determine the accuracy of euler's implicit method.

```
%Givens
LI = 10;
c0I = 3;
DI = 0.008;
tfI = 100;
dxI = 0.1;
dtI = 0.01;

%Grid parameters
NI = LI / dxI;
TI = tfI / dtI;

%Courant Number
CourantI = DI * dtI/dxI^2;

%\Concentration matrix
cI = zeros(NI, TI);
```

```

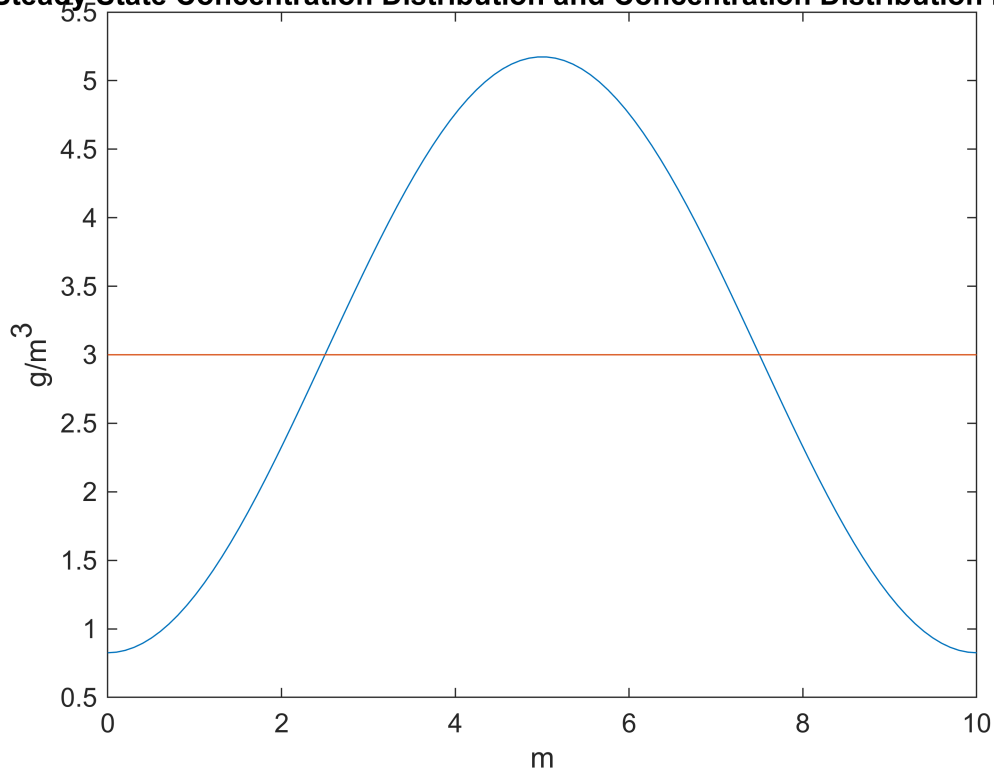
%Initial condition
xI = linspace(0, LI, NI)';
cI(:, 1) = c0I * (1 - cos((2*pi*xI)/LI));

% Solve the diffusion equation numerically using Euler's Implicit Method
for jI = 1:TI-1
    AI = diag(1 + 2*CourantI*ones(NI, 1)) - CourantI * diag(ones(NI-1, 1), 1) - CourantI * diag
    bI = cI(:, jI); %RHS
    bI(1) = bI(1) + CourantI * cI(2, jI); %BC @ Left Edge
    bI(NI) = bI(NI) + CourantI * cI(NI-1, jI); %BC @ Right Edge
    cI(:, jI+1) = AI\bI; %Solve linear system
end

%Plots
figure;
plot(xI, cI(:, TI));
xlabel('m');
ylabel('g/m^3');
title('Steady State Concentration Distribution and Concentration Distribution Implicit');
%Steady State @ 3 proven through limits on paper
c_steadyI = 3 * ones(NI, 1);
hold on;
plot(xI, c_steadyI);
hold off;

```

Steady State Concentration Distribution and Concentration Distribution Implici



```

%Analytical v Numerical

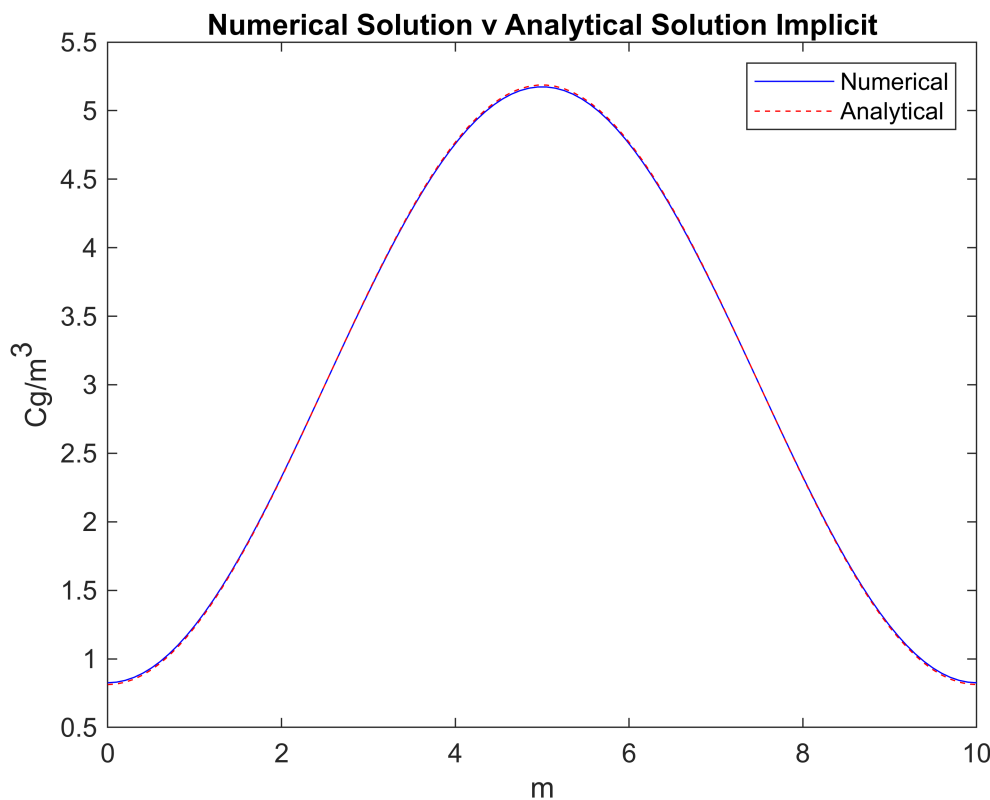
```



```

xNVA = linspace(0, LI, 100)';
canalyticalE = c0I * (1 - cos(2*pi*xNVA / LI).*exp(-(4*pi^2*DI*tfI)/(LI^2)));
figure;
plot(xI, cI(:, TI), 'b-');
hold on
plot(xNVA, canalyticalE, 'r--');
xlabel('m');
ylabel('Cg/m^3');
title('Numerical Solution v Analytical Solution Implicit');
legend('Numerical', 'Analytical');

```



Solutions for Both: I picked the courant number by comparing the time with the square of the spatial grid size, and multiplying it by the diffusion coefficient, the number ensures that solution is stable and convergent. I picked the highest spatial grid size (Which will make it more accurate due to more iterations) I could on my computer without well setting my laptop on fire from too much computing power. I tried to make it higher and Matlab requested 75 gb of ram which I do not have.

Results: I find that the numerical solution from the implicit method is more accurate compared to the analytical/ true solution than the explicit method. Therefore for this problem, I recommend using the implicit method to gain a more accurate numerical solution through a Matlab program.